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LoD (T2MP)

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Theory Recap - LoD

We formally define LoD as follows

$$\mathbf{LoD} = \langle \mathbf{ETG}, \models_{\mathbf{LoD}} \rangle$$

with

$$\mathbf{ETG} = \langle \mathbf{L}_{\mathbf{LoD}}, \mathbf{D}, \mathbf{I}_{\mathbf{LoD}} \rangle$$

When no confusion arises, we drop the subscripts.

Theory Recap - LoD Domain

Definition (Domain)

$$D = \langle \{C\}, \{R\} \rangle$$

where:

- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, where $u_1 \in U$, with $U = \{u\}$ is the **universe** of D
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of units, for some m , with $C_i \subseteq U$
- $\{R\}$ is a set of **binary relations** $R_1, \dots, R_p$ between units, for some p , with $R_i \subseteq U \times U$

Observation (Domain of interpretation). The units in U are not part of the domain of interpretation of LoD. The alphabet does not mention them.

Theory Recap - Classes

Definition (Class)

$$\{C\} = E_T \cup D_T$$

where:

- $U = \{u\}$ is the **universe of interpretation**;
- $E = \{e\} \subseteq \{u\} = U$ is the **entity universe**;
- $V = \{v\} \subseteq \{u\} = U$ is the **value universe**;
- $\{e\}$ and $\{v\}$ are disjoint.
- $E_T = \{E_i\}$ is a **set of etypes** E_i , with $E_i = \{e\} \subseteq E$
- $D_T = \{D_i\}$ is a **set of dtypes** D_i , with $D_i = \{v\} \subseteq V$

Observation (LoD classes). The same classes as LoE, but with no reference to the entities e and values v in U .

Theory Recap - Binary relations

Definition (Binary relation)

$$\{R\} = O_R \cup A_R$$

where:

- $O_R = \{O_i\}$ is a **set of object properties** O_i , with $O_i \subseteq E_k \times E_j$
- $A_R = \{A_i\}$ is a **set of attributes** A_i , with $A_i \subseteq E_k \times D_j$

Observation (Object and data relations). The same relations as LoE.

Theory Recap – BNF Language of Etype Percepts

$$\langle p_T \rangle ::= \langle \text{etype} \rangle \mid \langle \text{dtype} \rangle \mid \top \mid \perp$$
$$\begin{aligned} \langle \text{etype} \rangle &::= \exists \langle \text{objProp} \rangle . \langle \text{etype} \rangle \mid \\ &\quad \exists \langle \text{dataProp} \rangle . \langle \text{dtype} \rangle \mid \\ &\quad \forall \langle \text{objProp} \rangle . \langle \text{etype} \rangle \mid \\ &\quad \forall \langle \text{dataProp} \rangle . \langle \text{dtype} \rangle \end{aligned}$$
$$\langle \text{etype} \rangle ::= E_1 \mid \dots \mid E_n$$
$$\langle \text{dtype} \rangle ::= D_1 \mid \dots \mid D_n$$
$$\langle \text{objProp} \rangle ::= O_1 \mid \dots \mid O_n$$
$$\langle \text{dataProp} \rangle ::= A_1 \mid \dots \mid A_n$$

where: $E_1, \dots, E_n, \dots, A_1, \dots, A_n$ are etypes which have the same extension as reference concepts.

Theory Recap – BNF Language of Composite Etype Percepts

$$\begin{aligned} \langle p_C \rangle & ::= \langle p_C \rangle \sqcap \langle p_C \rangle \mid \\ & \quad \langle p_C \rangle \sqcup \langle p_C \rangle \mid \\ & \quad \neg \langle p_C \rangle \end{aligned}$$

$$\langle p_C \rangle ::= \langle p_T \rangle$$

Notation (BNF). $\langle p_C \rangle$ is a nonterminal symbol and it stands for a p_C percept. $\langle p_T \rangle$ is an L_C terminal symbol and stands for an L_T percept. See above how the BNF of L_T expands to a LoD terminal symbol.

Observation (BNF). This BNF does allow the iterative application of the formation rules. It allows to generate percepts of any depth.

Observation (Composite etypes and world models). Composite etypes cannot be represented in a world model.

Theory Recap – BNF for Descriptions

$$\langle a_{\text{LoD}} \rangle ::= \langle p_C \rangle \sqsubseteq \langle p_C \rangle \mid \langle p_C \rangle \equiv \langle p_C \rangle$$

where:

- $\langle p_C \rangle$ is a composite etype percept.
- $\langle a_{\text{LoD}} \rangle$ is a LoD description, an assertion involving two composite etype percepts.

Terminology (LoD Description). A **LoD description** describes how the extensions of two composite etype percepts correlate. It is a constraint which reflects back into the component etypes. We call the first a **subsumption** (description) and the second an **equivalence** (description).

Terminology (Subsumption). $\sqsubseteq$ is a **subsumption relation**. $p_1 \sqsubseteq p_2$ is to be read as p_1 is **subsumed by** p_2 , or, vice versa that p_2 **subsumes** p_1 .

Terminology (Equivalence). $\equiv$ is an **equivalence relation**. We have

$$p_1 \equiv p_2 \text{ if and only if } p_1 \sqsubseteq p_2 \text{ and } p_2 \sqsubseteq p_1$$

Theory Recap – BNF for Definitions

$$\begin{aligned} \langle a_{\text{LoD}} \rangle &::= \langle E \rangle \sqsubseteq \langle p_C \rangle \mid \langle E \rangle \equiv \langle p_C \rangle \\ \langle E \rangle &::= E_1 \mid \dots \mid E_n \end{aligned}$$

where:

- $\langle E \rangle$ is an atomic etype percept (an etype in L_T).
- $E_1 \mid \dots \mid E_n$ are (names of) etype percepts
- p_C is a composite etype percept.
- a_{LoD} is a **LoD definition**.

Terminology (LoD definition). A **LoD definition** is a LoD description that describes the extension of an atomic etype. It constrains the extension of $\langle E \rangle$. LoD definitions allow to introduce new etypes by defining their extension.

Terminology (Etype subsumption, etype equivalence). The first definition is an **etype subsumption**. The second is an **etype equivalence**. Equivalences allow to precisely define the extension of $\langle E \rangle$.

Theory Recap - LoD Interpretation function

Definition (Interpretation function, alphabet). The LoE interpretation function I is defined as

$$I : L \rightarrow D$$

where I is defined as

$$I = \langle I_T, I_P \rangle$$

where:

- $I_T = \langle I_E, I_D \rangle$, the Type interpretation function, is the pair of the etype and the dtypes interpretation function;
- $I_P = \langle I_O, I_A \rangle$, the Property interpretation function, is the pair of the object property and the attribute interpretation function

with:

$$I_E : \{E_i\} \rightarrow \{E_i\}$$

$$I_O : \{O_i\} \rightarrow \{O_i\}$$

$$I_D : \{D_i\} \rightarrow \{D_i\}$$

$$I_A : \{A_i\} \rightarrow \{A_i\}$$

Theory Recap - Interpretation of etype percepts

$I_T(T) = U$, with U the universe of interpretation

$I_T(\perp) = \emptyset$, with $\emptyset$ the empty set

$I_T(E_i) = E_i$

$I_T(D_i) = D_i$

$I_T(\exists P.T) = \{d \in U \mid \text{there is an } e \in U \text{ with } (d, e) \in I_T(P) \text{ and } e \in I_T(T)\}$

$I_T(\forall P.T) = \{d \in U \mid \text{for all } e \in U \text{ if } (d, e) \in I(P) \text{ then } e \in I_T(T)\}$

where I_T is the interpretation function of L_T

Observation (Interpretation function). For an intensional view of the interpretation functions for etypes, dtypes, object properties and attributes, follow what done with LoE.

Observation (Interpretation of nested etypes). It is sufficient to interpret the application of the second external quantifier to the etype built via the application of the first quantifier.

Theory Recap - Interpretation of composite etype percepts

$$I_C(p_1 \sqcap p_2) = I_C(p_1) \cap I_C(p_2)$$

$$I_C(p_1 \sqcup p_2) = I_C(p_1) \cup I_C(p_2)$$

$$I_C(\neg p_1) = U \setminus I_C(p_1)$$

$$I_C(p_T) = I_T(p_T)$$

$$I_T(p_T) = p_T$$

where:

- I_C is the interpretation function of L_C
- I_T is the interpretation function for L_T , the language of etype percepts.
- p_1, p_2 are composite etype percepts
- p_T (in *italic*) is (the name of an) etype percept denoting the domain percept p_T (not in *italic*)

Theory Recap - Interpretation of LoD descriptions

$$\begin{aligned} I(p_1 \sqsubseteq p_2) & \text{ iff } I_c(p_1) \subseteq I_c(p_2) \\ I(p_1 \equiv p_2) & \text{ iff } I_c(p_1) = I_c(p_2) \\ & \text{ iff } I_c(p_1) \subseteq I_c(p_2) \text{ and } I_c(p_2) \subseteq I_c(p_1) \end{aligned}$$

where:

- I is the interpretation function for L_{LoD}
- I_c is the interpretation function for L_c , the language of composite etype percepts

Observation. Differently from before, the interpretation function does not return a subset of the domain but a fact.

Theory Recap - Interpretation of LoD definitions

$$\begin{aligned} I(E \sqsubseteq p_2) & \quad \text{iff} \quad I_c(E) \subseteq I_c(p_2) \\ I(E \equiv p_2) & \quad \text{iff} \quad I_c(E) = I_c(p_2) \\ & \quad \text{iff} \quad I_c(E) \subseteq I_c(p_2) \text{ and } I_c(p_2) \subseteq I_c(E) \\ I_c(E) & \quad = \quad I_T(E) = E \end{aligned}$$

where:

- I is the interpretation function for L_{LoD}
- I_c is the interpretation function for L_c , the language of composite etype percepts
- I_T is the interpretation function for L_T , the language of composite etype percepts
- E is an etype, a subset of the universe of interpretation

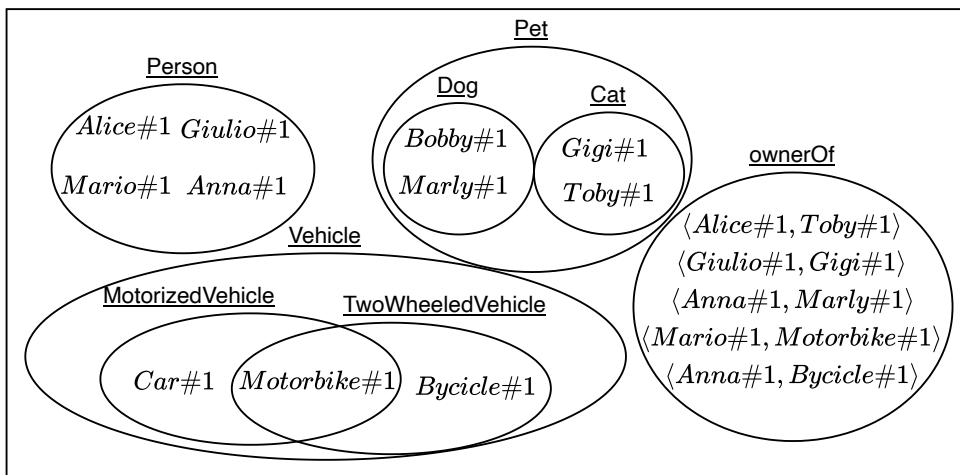
Observation. Again, differently from before, the interpretation function does not return a subset of the domain but a fact.

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Ex 1: Language of etype percepts

Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine the set of entities represented by the assertions in the **language of etype percepts**.



MotorizedVehicle {Car#1, Motorbike#1}

Pet {Bobby#1, Marly#1, Toby#1, Gigi#1}

$\exists \text{ownerOf.Pet}$ {Alice#1, Anna#1, Giulio#1}

$\forall \text{ownerOf.Cat}$ {Alice#1, Giulio#1, Car#1, Motorbike#1, Bicycle#1, Toby#1, Bobby#1, Gigi#1, Marly#1}

$\exists \text{ownerOf.T}$ {Alice#1, Anna#1, Giulio#1, Mario#1}

$\forall \text{ownerOf.}\perp$ {Car#1, Motorbike#1, Bicycle#1, Toby#1, Bobby#1, Gigi#1, Marly#1}

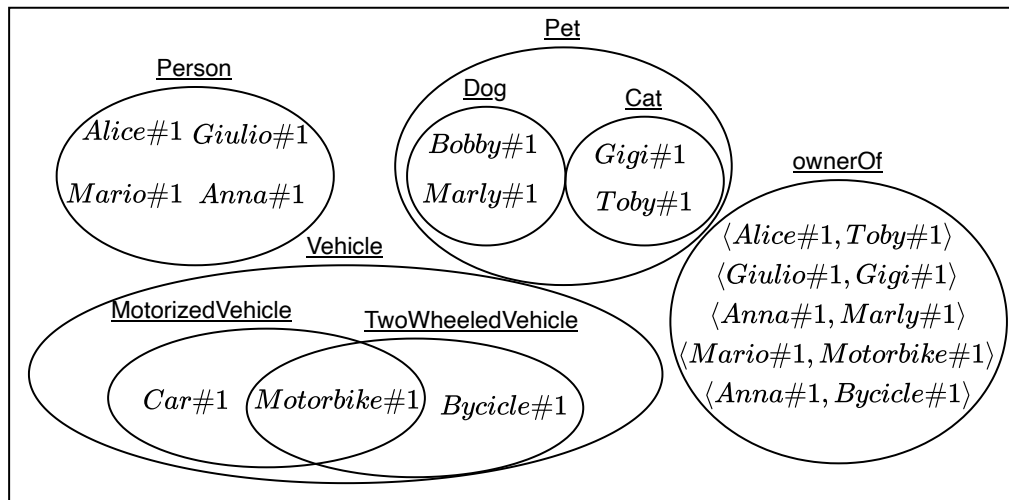
$\exists \text{ownerOf.}\perp$ {}

Ex 2: Language of composite etype percepts

Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine the set of entities represented by the assertions in the **language of composite etype percepts**.

$MotorizedVehicle \sqcup (Person \sqcap \forall ownerOf. Vehicle) \quad \{Mario\#1, Car\#1, Motorbike\#1\}$

$Person \sqcap \exists ownerOf. (Motorizedvehicle \sqcap TwoWheeledVehicle) \quad \{Mario\}$



$\neg Vehicle \sqcap \neg Dog \sqcap \forall ownerOf. \perp \quad \{Toby, Gigi\}$

$\exists ownerOf. (Pet \sqcap \neg Cat) \quad \{Anna\}$

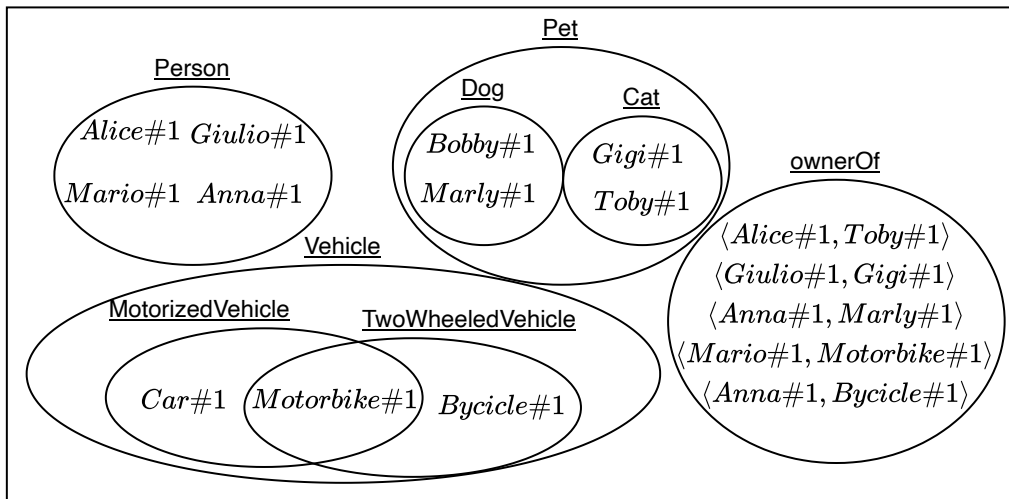
$Person \sqcap \neg \forall ownerOf. Cat \quad \{Anna, Mario\}$

$Person \sqcap \exists ownerOf. \neg Pet \quad \{Mario, Anna\}$

$Person \sqcap \exists ownerOf. (Vehicle \sqcap \neg TwoWheeledVehicle) \quad \{\}$

Ex 3: Language of Descriptions

Given the following intended model, assuming that the interpretation function is built by doing syntactic matching between the tokens used in the language and the names given to elements and sets of the model, determine if the following statements in **LoD** reflect correctly the intended model.



$MotorizedVehicle \sqsubseteq Vehicle$ True

$\exists ownerOf.Pet \sqsubseteq Person$ True

$\exists ownerOf.Pet \equiv Person$ False

$Dog \sqcup Cat \equiv Pet$ True

$\forall ownerOf.Vehicle \sqsubseteq Person$ False

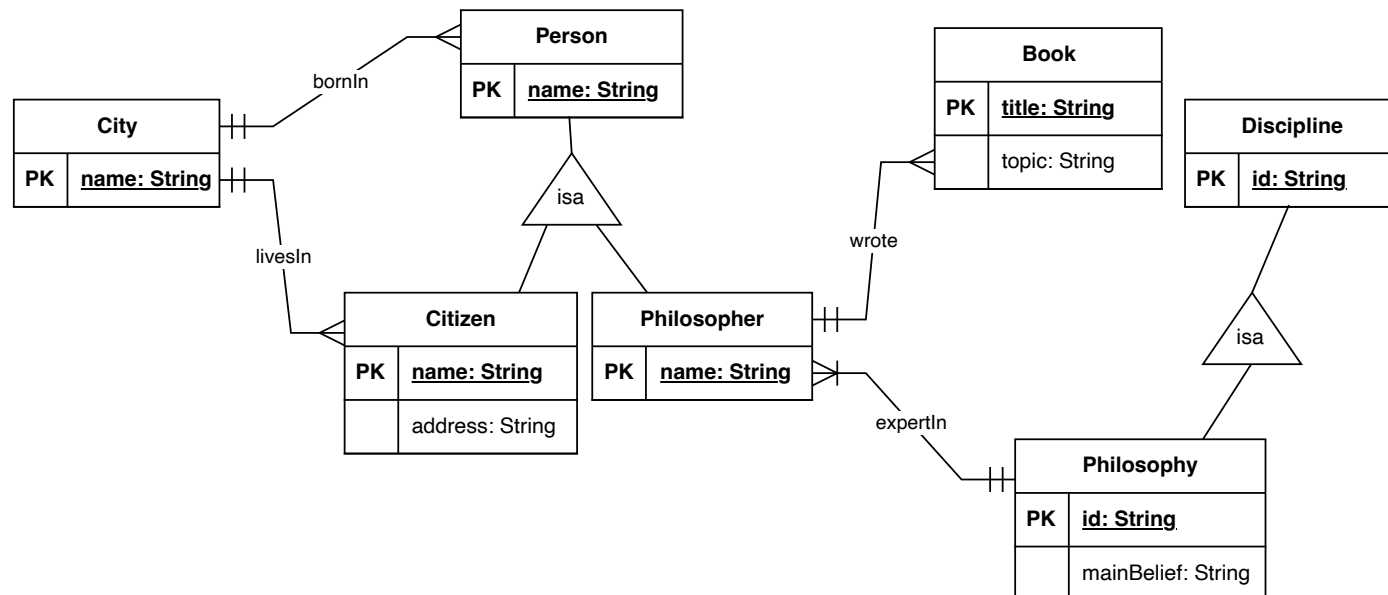
$\neg \forall ownerOf.Vehicle \sqsubseteq Person$ True

$Person \sqcap Pet \sqsubseteq \top$ True

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Example 1 - SF2F LoD



$City \sqsubseteq \exists name.String$

$Person \sqsubseteq \exists name.String \sqcap \exists bornIn.City$

$Book \sqsubseteq \exists title.String \sqcap \exists topic.String$

$Philosophy \sqsubseteq \exists mainBelief.String \sqcap Discipline$

$Citizen \equiv Person \sqcap \exists address.String \sqcap \exists livesIn.City$

$Philosopher \equiv Person \sqcap \exists expertIn.Philosophy \sqcap \forall wrote.Book$

This is a valid formalization of the EER diagram, but it allows for entities that are of etype $Person \sqcap City$ or similar.

To fix the issue we would need to add the following statements:

$Person \perp City$

$Person \perp Book$

$Person \perp Discipline$

$Book \perp Discipline$

$Book \perp City$

$Discipline \perp City$

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Ex 4 - Informal to formal

Determine if the following formalizations in the sub-languages of the Logic of Descriptions (Language of etype percepts: lep, Language of composite etype percepts: lecp, Language of descriptions: LoD) of the natural language sentences are correct.

- True • “The entities that live in a building” in “lecp” is $\exists \text{ livesIn. Building}$
- True • “The entities that talk only with whom doesn't have friends” in “lep” is $\forall \text{ talksTo. } (\forall \text{ hasFirend. } \perp)$
- False • “A chef is a person that works in a restaurant” in “lecp” is $\text{Chef} \sqsubseteq \text{Person} \sqcap \exists \text{ worksAt. Restaurant}$
- False • “Vegetarian recipes are exactly those that do not contain meat” in “LoD” is
 $\text{VegetarianRecipe} \equiv \text{Recipe} \sqcap \neg \forall \text{ contain. Meat}$
- False • “Smartphones are devices used to call and text” in “LoD” is $\text{Smartphone} \sqsubseteq \text{Device} \sqcap \exists \text{ usedFor. } (\text{Call} \sqcap \text{Text})$
- True • “Smartphones are devices used to call and text” in “LoD” is $\text{Smartphone} \sqsubseteq \text{Device} \sqcap \exists \text{ usedFor. } (\text{Call} \sqcup \text{Text})$
- True • “The vehicles that have a battery which cannot be recharged by pedaling” in “lecp” is
 $\text{Vehicle} \sqcap \exists \text{ hasBattery. } (\text{Battery} \sqcap \neg \exists \text{ chargedBy. Pedaling})$

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LoD Terminology

Definition (Uses). Let T be definitional TBox. Let $E \sqsubseteq p$ or $E \equiv p$ be a definition in T . Then we say that E **directly uses** E' , where E' is an atomic etype, if E' occurs in p . We say that E **uses** E' if E' occurs in the right hand side of a definition of an etype mentioned in p , and so on recursively.

Observation (Uses). “Uses” is defined as the transitive closure of directly uses.

Observation (Acyclic definitional TBox). A definitional TBox is **acyclic** if

- There is no type that uses itself, and
- There are no two definitions of the same etype

Observation 1 (acyclic TBox). The second requirement avoids any type using itself.

Observation 2 (acyclic TBox). An acyclic Tbox avoids the following situation:

$$E_1 \sqsubseteq \dots E_2 \dots, \quad E_2 \sqsubseteq \dots E_3 \dots, \quad \dots, \quad E_n \sqsubseteq \dots E_1 \dots$$

Definition (Terminology). A **Terminology** is an acyclic definitional TBox.

Observation (Terminology). Terminologies are key in the construction of human knowledge.

Ex 6 - Cyclic/Acyclic TBox

Check if the following TBoxes are Cyclic or Acyclic.

$\text{WristAccessory} \equiv \text{Accessory} \sqcap \exists \text{wearingOn.Wrist}$

$\text{NeckAccessory} \equiv \text{Accessory} \sqcap \exists \text{wearingOn.Neck}$

$\text{WristWatch} \equiv \text{Watch} \sqcap \text{WristAccessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

Acyclic

$\text{WristAccessory} \sqsubseteq \text{Accessory}$

$\text{NeckAccessory} \sqsubseteq \text{Accessory}$

$\text{WristAccessory} \sqsubseteq \neg \text{NeckAccessory}$

$\text{WristWatch} \equiv \text{Watch} \sqcap \text{WristAccessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

Acyclic

Ex 7 - Cyclic/Acyclic TBox

Check if the following TBoxes are Cyclic or Acyclic.

$\text{WristAccessory} \equiv \neg \text{NeckAccessory}$

$\text{NeckAccessory} \equiv \text{Accessory} \sqcap \exists \text{woreOn.Neck}$

$\text{WristWatch} \equiv \text{Watch} \sqcap \text{WristAccessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

$\text{Accessory} \equiv \text{WristWatch} \sqcup \text{Necklace}$

$\text{WristAccessory} \equiv \text{Accessory} \sqcap \neg \text{NeckAccessory}$

$\text{NeckAccessory} \equiv \text{Accessory} \sqcap \neg \text{WristAccessory}$

$\text{WristWatch} \equiv \text{Watch} \sqcap \text{WristAccessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

Cyclic

Because “Accessory” is defined in terms of “WristWatch” and “Necklace” which are in turn defined in terms of “NeckAccessory” which is defined in terms of “Accessory”.

Cyclic

Because “WristAccessory” and “NeckAccessory” are both defined in terms of the other.

Ex 8 - Terminology

Check if the following TBoxes are a Terminology.

$\text{WristAccessory} \equiv \text{Accessory} \sqcap \exists \text{wearingOn.Wrist}$

$\text{NeckAccessory} \equiv \text{Accessory} \sqcap \exists \text{wearingOn.Neck}$

$\text{WristWatch} \sqsubseteq \text{Watch} \sqcap \text{WristAccessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

Yes

$\text{WristAccessory} \equiv \text{Accessory} \sqcap \exists \text{wearingOn.Wrist}$

$\text{NeckAccessory} \equiv \text{Accessory} \sqcap \exists \text{wearingOn.Neck}$

$\text{WristAccessory} \sqsubseteq \neg \exists \text{wearingOn.Neck}$

$\text{WristWatch} \equiv \text{Watch} \sqcap \text{WristAccessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

No

Because it includes two definitions of “WristAccessory”

$\text{WristAccessory} \equiv \text{Accessory} \sqcap \forall \text{wearingOn.Wrist}$

$\text{NeckAccessory} \equiv \text{Accessory} \sqcap \forall \text{wearingOn.Neck}$

$\text{WristWatch} \equiv \text{Watch} \sqcap \exists \text{wearingOn.Wrist} \sqcap \neg \text{Accessory}$

$\text{Necklace} \equiv \text{NeckAccessory} \sqcap \forall \text{hasFunction.Decorative}$

$\text{Accessory} \equiv \exists \text{wearingOn.T} \sqcap \neg \text{WristWatch}$

No

Because it's cyclic.

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Logical entailment (Recall)

Definition (Logical entailment). Let $L_W = \langle W, \models_W \rangle$ be a world logic with $W = \langle L, D, I \rangle$ its world model. Let $\{M\} \subseteq D$ be a set of models of T with $T, \underline{I} \subseteq L$ and $a \in L$. Then we write

$$T \models_{\{M\}} \underline{I} \qquad (T \models_{\{M\}} a)$$

and say that T **logically entails** $\underline{I}$ (T **logically entails** a) with respect to $\{M\}$ if

$$\text{for all } M \in \{M\}, \text{ if } M \models T \text{ then } M \models \underline{I} \text{ (} M \models a \text{)}$$

Observation (Model filtering via logical entailment). Logical entailment implements reasoning in the following way: “Whatever we derive from T must be true in the same sets of “relevant” models $\{M\}$ as T . T acts as a filter which allows to discard all the irrelevant models.

Observation (Model filtering via selection of $\{M\}$). The choice of the models $\{M\}$ of T is arbitrary. And it also amounts to model filtering. It could be the only model of T , all models of T , a model selected because it is the intended model of T , and so on. If $M \not\models T$ then Logical entailment does not apply. T has the key role of selecting which models $M \in \{M\}$ must be considered.

Notation ($\{M\}$). When no confusion arises we write $\models$ instead of $\models_{\{M\}}$.

LoD world entailment

$$M \models p_1 \sqsubseteq p_2 \quad \text{iff} \quad I(p_1 \sqsubseteq p_2) \quad \text{iff} \quad I(p_1) \subseteq I(p_2)$$

$$M \models p_1 \equiv p_2 \quad \text{iff} \quad I(p_1 \equiv p_2) \quad \text{iff} \quad I(p_1) = I(p_2)$$

$$\text{iff} \quad I(p_1) \subseteq I(p_2) \text{ and } I(p_2) \subseteq I(p_1)$$

with $p_1, p_1 \in L_{\text{LoD}}$.



World entailment (extended question language)

$$M \models p_1 \sqsubseteq p_2 \quad \text{iff} \quad I(p_1 \sqsubseteq p_2) \quad \text{iff} \quad I(p_1) \subseteq I(p_2)$$

$$M \models p_1 \equiv p_2 \quad \text{iff} \quad I(p_1 \equiv p_2) \quad \text{iff} \quad I(p_1) = I(p_2) \\ \text{iff} \quad I(p_1) \subseteq I(p_2) \text{ and } I(p_2) \subseteq I(p_1)$$

$$M \models p_1 \perp p_2 \quad \text{iff} \quad I(p_1 \sqsubseteq \neg p_2) \quad \text{iff} \quad I(p_1) \cap I(p_2) \subseteq \emptyset$$

$$M \models p_1 \perp\!\!\!\perp p_2 \quad \text{iff} \quad I(p_1 \equiv \neg p_2) \quad \text{iff} \quad I(p_1) \cap I(p_2) \subseteq \emptyset \text{ and } I(p_1) \cup I(p_2) \subseteq U$$

with

- $p_1, p_2 \in L_{\text{LoD}}$;
- $p_1 \perp p_2$ a notational variant of $p_1 \sqsubseteq \neg p_2$.
- $p_1 \perp\!\!\!\perp p_2$ a notational variant of $p_1 \equiv \neg p_2$.

LoD – Reasoning problems

Observation (LoD reasoning problems). The four LoD core reasoning problems are:

- $T \models C$ Satisfiability with respect to a TBox T
- $T \models C \sqsubseteq D$ Subsumption with respect to a TBox T
- $T \models C \equiv D$ Equivalence with respect to a TBox T
- $T \models C \perp D$ Disjointness with respect to a TBox T

where T can also be empty.

Observation (Empty TBox). All LoD reasoning problems are logical entailment problems. When the TBox is empty, then LoD logical entailment reduces to LoD world entailment on the reference model(s).

Ex 9 - Reasoning

Given the TBox below, determine if the assertions on the right are satisfiable.

M	$City \sqsubseteq \exists name.String$	$M \models \text{Philosopher} \sqcup \text{Citizen} \sqsubseteq \text{Person}$	T
	$Person \sqsubseteq \exists name.String \sqcap \exists bornIn.City$	$M \models \text{City} \sqsubseteq \neg \text{Book}$	F
	$Book \sqsubseteq \exists title.String \sqcap \exists topic.String$	$M \models \text{Philosopher} \sqcap \text{Citizen} \sqsubseteq \text{Person}$	T
	$Philosophy \sqsubseteq \exists mainBelief.String \sqcap Discipline$	$M \models \text{Book} \sqcap \text{City} \sqsubseteq \perp$	F
	$Citizen \sqsubseteq \text{Person} \sqcap \exists address.String \sqcap \exists livesIn.City$	$M \models \text{Person} \sqcap \text{City} \sqsubseteq \top$	T
	$Philosopher \sqsubseteq \text{Person} \sqcap \exists expertIn.Philosophy \sqcap \forall wrote.Book$	$M \models \text{Philosopher} \sqsubseteq \exists bornIn.City \sqcup \exists address.String$	T
	$Person \sqperp City$		
	$Person \sqperp Book$		
	$Person \sqperp Discipline$		



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